

Grofers: Building a hyperlocal delivery service using Google Cloud

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ABOUT GROFERS

Google Cloud results

- Scales from 10,000 orders to 100,000 orders a day
- Increases the average number of orders per driver from five to nearly 15
- Maintains 100 percent uptime

India-based Grofers (<https://grofers.com>) is a hyperlocal mobile platform that delivers groceries, fruits and vegetables, cosmetics, and electronics. Customers use Grofers' mobile app to shop for products from local stores in 17 cities in India.

In 2015, the number of deliveries was growing at the rate of 10 times a month. The delivery system struggled under heavy traffic spiked on weekends, its d



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Industries: Technology

Location: India

crashed. Fulfilling and delivering manual processes such as telephoning customers and drivers needed a solution that would allow them to scale and improve its delivery process.

[App Engine \(https://cloud.google.com/appengine/\)](https://cloud.google.com/appengine/)

[Cloud Endpoints](https://cloud.google.com/endpoints/)

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"Our customer experience and reputation took a hit because our logistics system couldn't keep up with the number of orders being placed. We chose Google Cloud because it scales instantly from one thousand orders to one million orders while offering tools for quickly developing powerful apps," says Rohit Prakash, associate vice president of engineering at Grofers.

Building an entire back end in two weeks

When Grofers launched in 2013, the logistics system was entirely manual. Staff at the headquarters called delivery drivers with order details and delivery addresses. Drivers then decided what order to pick up goods from local merchants. With a small number of customers and orders, that wasn't a problem. But the manual system couldn't handle growth. Grofers recognized it had to automate the system.

To do that, it turned to [Google Cloud](https://cloud.google.com) (https://cloud.google.com). Grofers built a new back end for its delivery system using [App Engine](https://cloud.google.com/appengine) (https://cloud.google.com/appengine). When an order arrives, the new system looks at the status of all drivers — including their current location, the order he's completing, and the pickup and delivery locations on the route. Based on that information, the system calculates the optimal driver for each

order and sends the driver the order via a mobile app.

The system also takes advantage of Cloud Endpoints (<https://cloud.google.com/endpoints/>), which generates libraries for Javascript, iOS, and Android. Grofers used Cloud Endpoints to create mobile apps to communicate with its delivery staff.

“In only two weeks, I built the entire back end, including the mobile apps, using Google App Engine. It’s extremely easy to use and has no learning curve. I didn’t have to hire dedicated app experts, which saved us time and money,” Prakash says.

Scaling fast, dramatically improving delivery metrics

With the new backend and mobile apps, Grofers can scale quickly to keep up with its rapid growth. Grofers now delivers 70,000 orders per day, up from 10,000 before the Google Cloud deployment, and has had 100 percent uptime. Each driver handles an average of 15 orders per day, compared to only five previously, letting Grofers handle more work with less staff. All this has been accomplished at very low cost: Prakash says that Google Cloud costs account for only two percent of the company’s infrastructure spending. Without Google Cloud, Prakash says the back end would be far more expensive and account for up to 50 percent of the company’s infrastructure expenses.

“With Google Cloud, we had a very small learning curve and no setup time. We now have a low-cost

platform that can handle 70,000 orders per day and more no matter how large we grow,” Prakash says.